Liam C Pendleton, MSc

Quantitative Biologist

Seattle, WA • (734) 925-4389 • liamcpendleton@gmail.com • liampendleton.github.io

Summary

Avian conservation scientist with 8+ years of experience leveraging analytical and field methods to inform conservation decisions. I use long-term ecological monitoring, predictive models, and machine learning methods to monitor avian population trends and behavior. My expertise includes frequentist and Bayesian statistical modeling, data management, spatial and machine learning analytical methods, team supervision, and cross sector collaboration.

Professional Experience

Washington Cooperative Fish and Wildlife Research Unit — Seattle, WA

Research Assistant — 07/2021 - 06/2024

Collaborated with state, federal and community science partners to design and implement a long-term seabird monitoring program at Protection Island, WA. Used Bayesian modeling, GIS, and machine learning techniques to inform population trends and behavior of two seabird species monitored by the Puget Sound Partnership as indicators of Puget Sound restoration goals.

- Managed logistics for field operations including equipment procurement, scheduling, and resource allocation
- Organized and trained field teams of 5+ members annually, and directed 20+ field days per season to collect spatial and reproductive data
- Built hierarchical models in a Bayesian framework using R and JAGS to estimate effects of biological and physical processes on reproductive success of pigeon guillemots
- Used hidden Markov modeling in R to identify state-specific behaviors and foraging habitat of of pigeon guillemots and rhinoceros auklets
- Organized and participated in a various scientific communication events targeting audiences of varying levels of expertise

Bird Laboratory, Eastern Michigan University — Ypsilanti, MI

Research Assistant — 08/2019 - 05/2021

Conducted a spatial analysis of habitat use by a protected songbird species, using R-based modeling and spatial data to define and identify core habitat. Highlighted potential conservation priorities across 5,000+ hectares of habitat.

- Managed a large-scale spatial dataset describing habitat usage across 5,000+ hectares; validated and standardized data to support analyses
- Used kernel density estimations and Brownian bridge movement models in R to describe core habitats and foraging patterns
- Analyzed acoustic files using Audacity to assess energy expenditure of breeding and non-breeding red crossbills
- Led logistical planning for a multi-week field data collection trip along the US west coast, including travel coordination, equipment procurement, and scheduling for team members
- Collected morphometric and physiological data from 12 birds, adding ~20% more data points to existing datasets

Aquatic Geochemistry Laboratory, University of Michigan — Ann Arbor, MI

Data Analyst — 02/2020 - 09/2020

Identified primary catalysts of methane-producing reactions in thawed Alaskan permafrost carbon conversion while streamlining research workflow through statistical modeling, version control, and reporting.

- Used R and Excel to assess annual variation in 15+ chemical properties of 50+ Alaskan permafrost samples, identifying sunlight, bacterial activity and iron concentration as significant contributors to methane production
- Provided weekly data summaries to the principal investigator, resulting in improved efficiency of ongoing lab projects and project workflow
- Used GitHub to track and manage 5 iterations of analysis code

Bird Center of Washtenaw County — Ann Arbor, MI

Data Coordinator & Supervisor — 05/2017- 11/2019

Oversaw patient data management and tracking, leading database optimization, digital system implementation, and data-driven decision-making to improve operational efficiency and patient care.

- Hired, supervised and trained 20+ interns and volunteers; created training materials and provided mentorship and professional development
- Coordinated annual fundraising and community outreach events, securing $30,000+ in donations annually
- Developed event and operational budgets; forecasted expenses, tracked costs, and oversaw equipment procurement
- Collaborated with board members and external partners to build partnerships supporting the Bird Center's growth
- Managed a database containing 3,000+ patient records, overseeing data entry and retrieval processes
- Wrote annual reports describing clinic statistics including seasonal trends in patient circumstances, rehab success rates, and species treated

Detroit Audubon Society — Detroit, MI

Field Technician — 05/2018 - 08/2019

- Monitored 30+ shorebird nests across multiple sites, tracking and reporting both sites and environmental conditions
- Collected and analyzed aquatic condition data (e.g., water levels, temperature, vegetation types)
- Assisted in restoring 3 acres of native prairie habitat using vegetation surveys and habitat assessments, increasing nesting and foraging opportunities for native bird and pollinator species
- Led community outreach efforts, engaging with 50+ residents to promote habitat conservation and increase awareness of native wildlife

Education

University of Washington | 2024

M.S., Aquatic and Fishery Sciences

Eastern Michigan University | 2021

B.S., Biology

Washtenaw Community College | 2018

A.A., Liberal Arts

Volunteering

Pacific Seabird Group — Seattle, WA

Organizer — 11/2023 - 03/2024

- Designed and ran a professional development panel targeted at early career scientists. Contacted and organized panelists from varying professional backgrounds, wrote starter questions, and facilitated dialogue between panelists and audience.
- Assisted in organizing a professional mentoring event. Contacted and confirmed mentors, and paired with student mentees.

Salish Sea Guillemot Network — Seattle, WA

Field Technician — 05/2022 - 08/2023

- Monitored foraging activity of pigeon guillemots at various sites around Puget Sound, WA. Identified prey species, nest locations, and provisioning frequency.

Ann Arbor Natural Area Preservation — Ann Arbor, MI

Field Technician — 05/2019 - 08/2020

- Conducted point counts and vegetation surveys at 10+ sites around Ann Arbor, MI to assess songbird abundance and diversity following habitat restoration initiatives.